

HR Analytics: Job Change of Data Scientists

1. Introduction

This project focuses on a binary classification problem using the HR Analytics: Job Change of Data Scientists dataset from Kaggle. The goal of the project is to predict whether a candidate is likely to look for a job change based on demographic, educational, and professional information.

The target variable used in this project is target, where:

- 0 = candidate is not looking for a job change
- 1 = candidate is likely to look for a job change

The project builds on the preprocessing workflow developed during the midterm project and extends it by training, validating, and comparing multiple machine learning classification models.

2. Dataset and Preprocessing

The dataset contains both numerical and categorical features, including:

- city
- education level
- relevant experience
- company size
- company type
- training hours
- years of experience
- last new job

The original raw dataset (aug_train.csv) was used for the final project.

The preprocessing logic from the midterm project was reused and adapted for the final machine learning workflow. The following preprocessing steps were applied:

- handling missing values using median and most frequent imputation
- ordinal encoding for ordered categorical variables
- one-hot encoding for nominal categorical variables
- feature scaling using StandardScaler

- feature engineering for experience and training-related variables
- log transformation of skewed numerical features

The dataset was split into:

- 70% training data
- 15% validation data
- 15% testing data

Stratified splitting was used to preserve class distribution across all datasets.

3. Machine Learning Models

The following classification models were trained and evaluated:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier
- Gradient Boosting Classifier
- K-Nearest Neighbors Classifier
- Support Vector Classifier (SVC)

The following evaluation metrics were used:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

ROC-AUC was considered one of the most important metrics because the dataset is moderately imbalanced.

4. Ensemble Models

In addition to the individual classification models, two ensemble approaches were evaluated:

Voting Ensemble

A soft voting classifier was created using the top three performing models based on validation ROC-AUC scores.

Bayesian Ensemble

A Bayesian Ridge ensemble model was trained using the prediction probabilities generated by the top-performing classifiers.

These ensemble methods were used to compare whether combined predictions could outperform the best individual model.

5. Results and Model Comparison

	Model	Dataset	Accuracy	Precision	Recall	F1 Score	ROC-AUC
7	Gradient Boosting	Test	0.7756	0.5661	0.4246	0.4852	0.7891
13	Voting Ensemble	Test	0.7791	0.5595	0.5321	0.5455	0.7873
5	Random Forest	Test	0.7735	0.5455	0.5447	0.5451	0.7837
15	Bayesian Ensemble	Test	0.7582	0.5175	0.4330	0.4715	0.7412
1	Logistic Regression	Test	0.7074	0.4405	0.6466	0.5241	0.7334
9	K-Nearest Neighbors	Test	0.7491	0.4958	0.4106	0.4492	0.7132
11	Support Vector Classifier	Test	0.7220	0.4446	0.4651	0.4546	0.6852
3	Decision Tree	Test	0.7227	0.4444	0.4525	0.4484	0.6320
6	Gradient Boosting	Validation	0.7888	0.6054	0.4407	0.5101	0.8061
12	Voting Ensemble	Validation	0.7898	0.5879	0.5272	0.5559	0.7992
4	Random Forest	Validation	0.7878	0.5790	0.5467	0.5624	0.7950
14	Bayesian Ensemble	Validation	0.7672	0.5424	0.4282	0.4786	0.7455
0	Logistic Regression	Validation	0.7122	0.4462	0.6360	0.5244	0.7321
8	K-Nearest Neighbors	Validation	0.7512	0.5018	0.3849	0.4357	0.7210
10	Support Vector Classifier	Validation	0.7370	0.4734	0.4840	0.4786	0.6923
2	Decision Tree	Validation	0.7161	0.4298	0.4226	0.4262	0.6181

Among all evaluated models, the Gradient Boosting Classifier achieved the best overall ROC-AUC performance on both the validation and testing datasets.

Key results:

- Validation ROC-AUC: 0.8061
- Test ROC-AUC: 0.7891

The Voting Ensemble also produced strong and competitive performance, especially in balancing precision and recall. However, it did not outperform the Gradient Boosting model in overall ROC-AUC.

The Bayesian Ensemble achieved lower performance compared to both the best individual model and the Voting Ensemble.

6. Conclusion

This project demonstrates a complete machine learning classification workflow using the HR Analytics Job Change dataset.

The preprocessing strategy developed during the midterm project was successfully reused and extended for machine learning model training and evaluation. Multiple classification algorithms were compared using validation and testing datasets, and ensemble methods were also evaluated. The results showed that the Gradient Boosting Classifier achieved the best overall performance for predicting whether a candidate is likely to look for a job change.

This project highlights the importance of proper preprocessing, feature engineering, model comparison, and evaluation using multiple performance metrics in real-world machine learning tasks.